

EnergyPilot

A home energy planner that learns the family's routine, forecasts demand from the real weather, schedules the heat pump and appliances for the lowest cost, and sizes a solar roof.

Team: [Team name] Members: [Names] Demo video: [link]

Forecast date shown: **Mon 28 Sep 2026** Location: **Katowice, Poland**

Q1 · Next 24
hours

**18.4
kWh**

likely 15–20 kWh ·
7 days: 125.5 kWh

Q2 · Peak hours

**06:00 ·
18:00**

short peaks up to
7.1 kW

Q3 · Plan saves

–23 %

€4.43 → €3.39
today, same
comfort

Q4 · 5 kWp solar

7.7 yrs

payback · 6.8 yrs
with smart habits

EnergyPilot answers the four questions of the challenge for Marek, Ania, Kuba and Zosia in one application. It forecasts the next 24 hours, 3 days and 7 days, explains when and why demand peaks, moves flexible loads and heat-pump operation to cheaper hours without reducing comfort, and shows what a rooftop PV system would change in energy, money and payback.

The key idea is that the house itself can store energy. The heat pump uses 62 % of the family's electricity; warming the older house slightly while power is cheap and letting it rest in expensive hours moves heating energy out of the peak without any new hardware. Combined with a forecast that knows the family calendar (Marek's shift rota, Ania's home-office days, school), this turns a busy, irregular household into a predictable and optimisable one.

1 · THE PROBLEM

A busy family with an irregular week

Marek and Ania live with Kuba (14) and Zosia (10) in an older detached house in Silesia, heated by a heat pump. Their energy use is hard to predict and expensive at the wrong moments:

- **Marek works three rotating shifts** (morning, afternoon, night), so he leaves, returns and sleeps at different times every week.
- **Ania works hybrid**: at home on Tuesdays and Thursdays and on some other days, in the office otherwise.
- **The children** follow school days, holidays and afternoon activities, then use TVs, consoles and computers.
- **Mornings and evenings are crowded**: kettle, coffee machine, oven, hob, washing machine and dishwasher often run together, just as the electricity price is highest (€0.40/kWh from 17:00 to 22:00).
- **The heat pump** dominates winter consumption and follows the outdoor temperature.

The family does not want to change its lifestyle. It wants to know **how much energy it will need, when demand will be highest and why, what small changes would lower the bill, and whether solar panels would pay off.**

2 · OUR SOLUTION

EnergyPilot in four steps

STEP 1 Learn the family's habits

A year of realistic hourly consumption per appliance, built from the family calendar, the brief's appliance table, a heat-pump model and real Katowice weather.

STEP 2 Forecast demand

The best of 8 tested models predicts every hour of the next 24 hours, 3 days and 7 days from the weather forecast and who will be at home.

STEP 3 Optimise the cost

An optimiser moves the washing machine and dishwasher to cheap hours and pre-heats the house before the peak, inside the family's comfort band.

STEP 4 Simulate solar panels

Any size from 0 to 15 kWp: production, share of demand covered, grid reduction, savings and payback with and without smart habits.

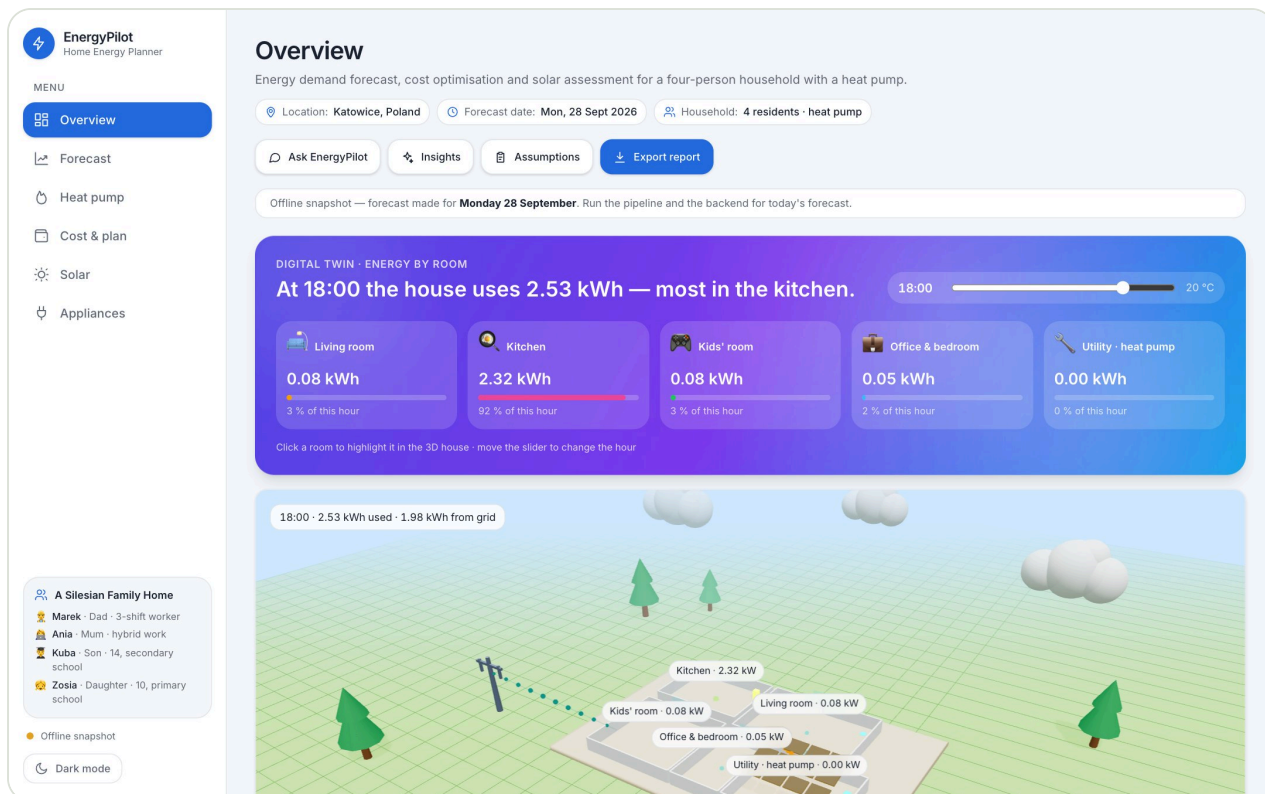


Figure 1. The Overview opens with the digital twin of the house: at 18:00 the house uses 2.53 kWh, 92 % of it in the kitchen. Each room tile is clickable and moves the 3D camera to that room.

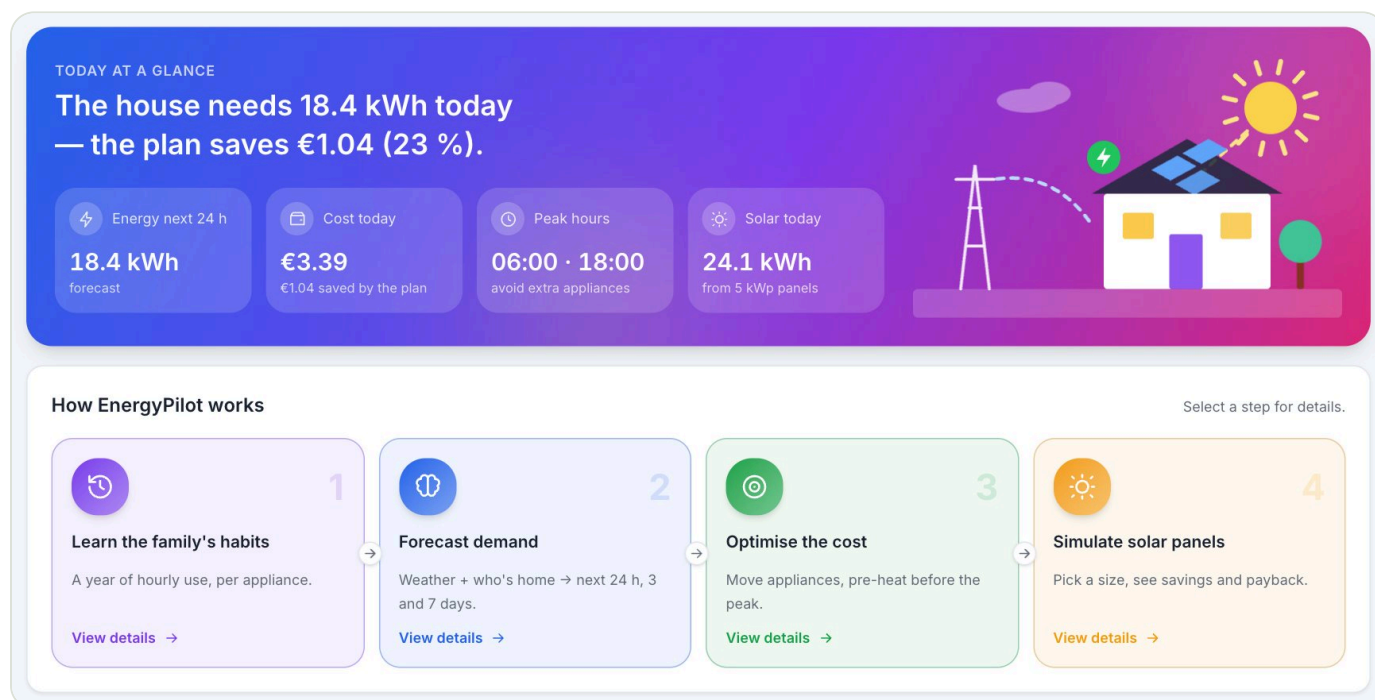


Figure 2. "Today at a glance" and the four steps. Every step opens the page that shows it in detail.

Q1–Q4, answered with the dashboard

All figures below come from the running system for the forecast of Monday 28 September 2026 (5 kWp scenario, "Comfort" level).

Q1 · EXPECTED DEMAND

What energy demand can be expected over the next 24 hours, 3 days and 7 days?

Period	Forecast	Likely range	Average outdoor
Next 24 hours	18.4 kWh	15.2–20.2 kWh	16.1 °C
Next 3 days	46.8 kWh	42.3–46.8 kWh	16.1 °C
Next 7 days	125.5 kWh	110.3–125.5 kWh	14.8 °C

Demand rises through the week as the weather cools: about 14–15 kWh on the mild Tuesday and Wednesday, about 20–21 kWh from Friday, because the heat pump works more. Heating is 30 % of the next 24 hours.



Figure 3. Forecast page: hourly forecast with the likely range (shaded), the busiest hours (orange) and the real outdoor temperature forecast (dashed line). The switch changes between 24 h, 3 days and 7 days.

Q2 · PEAK HOURS

During which hours will energy demand be highest?

- **06:00 (3.07 kWh):** the heat pump recovers the night set-back while breakfast is made and the children get ready.
- **18:00–19:00 (2.54 and 1.67 kWh):** dinner (oven and hob), screens, lighting and heating with the whole family at home.
- Inside an hour, **short power peaks reach up to 7.1 kW** when the hob, oven and kettle run together; a separate model forecasts these 1-minute peaks.
- Over 7 days, the three highest hours are at 06:00 on Friday, Sunday and Monday, the colder mornings.

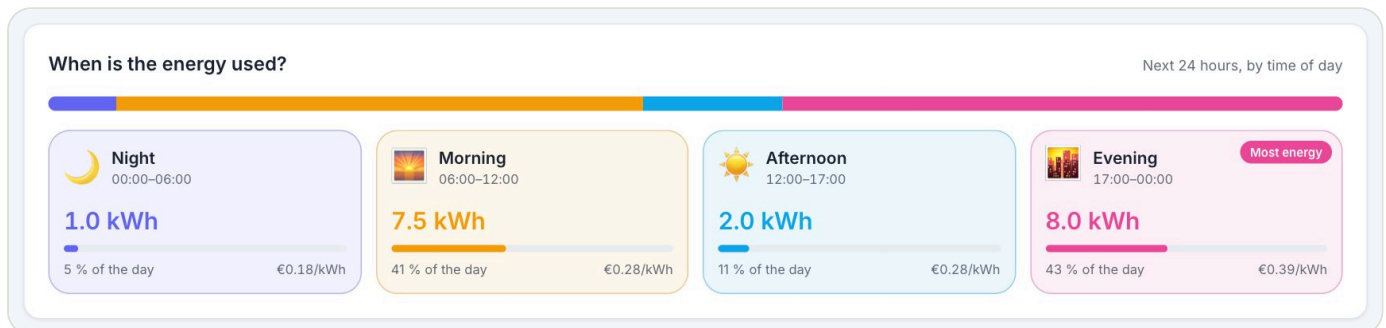


Figure 4. The evening (43 %) and morning (41 %) use most of the day's energy; the night, when electricity costs €0.18, only 5 %.

Q3 · HABIT CHANGES

What changes in habits could optimise the use of renewable energy and the grid?

Change	Why	Saving today
Washing machine 06:00 → 10:00	out of the morning peak, into sunny hours	€0.22
Dishwasher 20:00 → 00:00	from €0.40 to the €0.18 night price	€0.22
Heat pump pre-heats at 04:00, 05:00, 09:00, 10:00 and rests at 06:00–07:00	the house stores warmth while power is cheap; indoor stays 20.5–22.0 °C	€0.24
Solar and timing together	moved loads use more of the own solar power	€0.36
Total	€4.43 → €3.39, same comfort	€1.04 (23 %)

The highest grid hour falls from 3.83 to 1.98 kWh. A comfort slider lets the family trade warmth for money: from €0.69 a day (warmest band) to €1.59 a day (widest band). The week calendar marks the cheapest, calmest hour of every day.

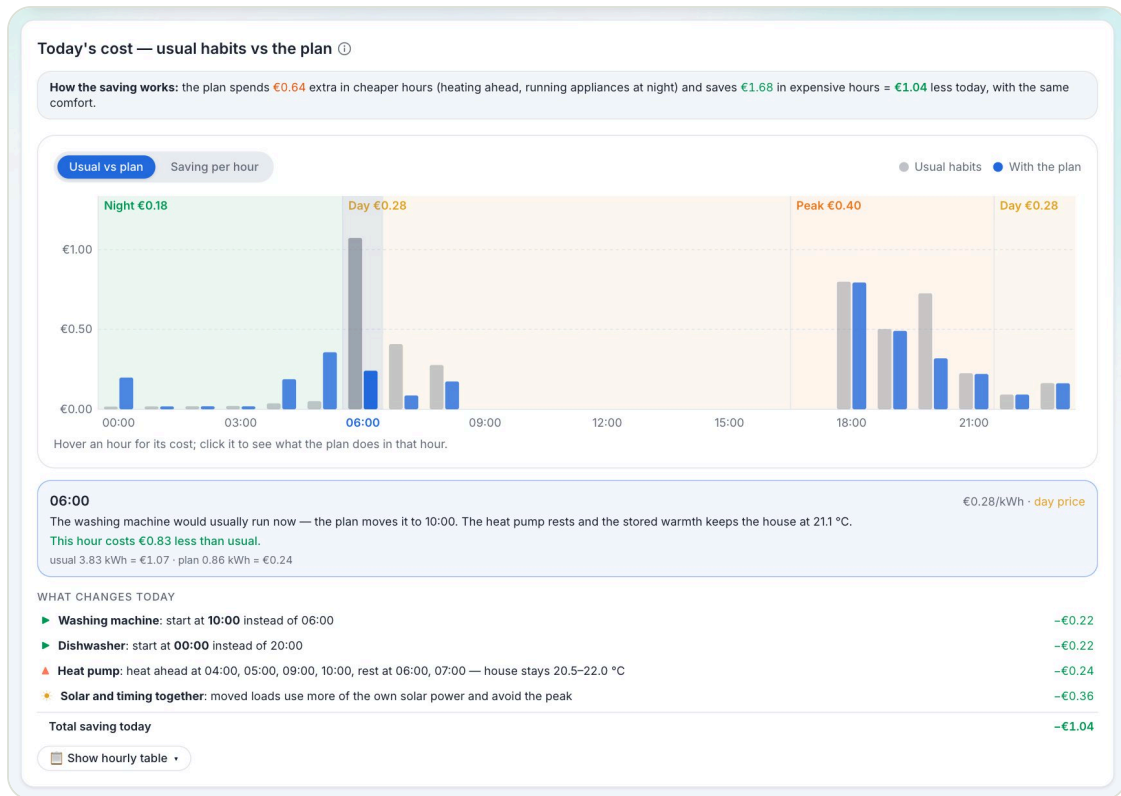


Figure 5. Cost & plan: each hour's cost with usual habits (grey) and with the plan (blue) over the tariff zones. Clicking an hour explains what the plan does in it.

Q4 · SOLAR INVESTMENT

How would a solar installation affect grid purchases, costs and payback?

For a 5 kWp system ($\approx$ 12 panels, €6,500): **5,374 kWh a year**, **21 % of the household's demand** covered, **2,296 kWh less** bought from the grid and a net saving of **€848 a year**. It pays back in **7.7 years** with current habits (A) and **6.8 years** when the washing machine and dishwasher run in sunny hours (B). Full comparison in section 7.

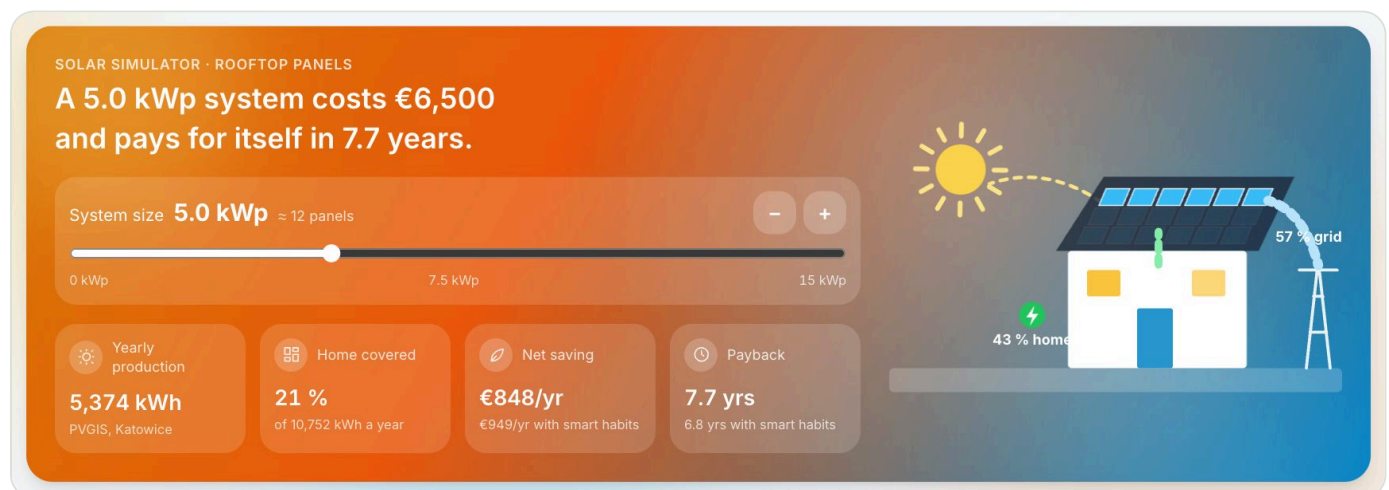


Figure 6. Solar simulator: the slider sets any size from 0 to 15 kWp; panels appear on the roof and the energy flows to the house and the grid update.

Everything the brief asks for

Required outcome (brief, section 08)	How EnergyPilot delivers it	Where
1. Historical hourly profile, $\geq$ 30 consecutive days	362 days, 8,688 hours, per appliance	<code>data/history.csv</code>
2. Forecasting model	8 models compared, best one chosen, calibrated	<code>train_model.py</code>
3. Hourly forecast for the next 24 hours	From 00:00 today, with likely range	Forecast page
4. Forecasts for the next 3 and 7 days	168 hours in the same run	Forecast page switch
5. Application presenting the results	Web dashboard, desktop to phone, offline one-file version	All pages
6. Hours with the highest expected demand	Top hours per period, short 1-minute peaks	Forecast, Overview
7. Brief explanation of changes in consumption	"Why this forecast" and day-by-day reasons	Forecast page
8. Renewable energy simulator	0–15 kWp, production, coverage, grid reduction, savings, payback A and B	Solar page

The application also shows everything section 3 of the brief lists: hourly consumption, the period total, the outdoor temperature forecast, the busiest hours and a short explanation.

A realistic year, built bottom-up

The family is fictional, so no meter data exists; the brief asks teams to *prepare* a realistic profile. We simulate every hour from 1 October 2025 to 27 September 2026 from the family's routine and the brief's appliance table, driven by the real weather for Katowice.

OCCUPANCY

Family calendar

Each person is away, awake or asleep every hour: Marek's weekly rotating shifts, Ania's home-office days, school, football and dance, holidays. Random choices are repeatable (seeded per day).

VARIABILITY

Unplanned events

Weekend outings, evenings out, sick days and trips are added on a second layer that the forecast never sees, so the model is tested fairly.

APPLIANCES

16 appliances, brief ranges

Meals drive cooking; screens run only when their users are home and awake; washing cycles need an adult at home; lighting follows real darkness. Minute-level loads give the short peaks.

WEATHER

Heat-pump physics

A thermal model of an older house (heat loss 0.25 kW/°C) with real Open-Meteo temperature and sun, a COP from 3.6 at 7 °C to 2.4 at -7 °C, and a gradual summer switch-off between 13 °C and 17 °C.

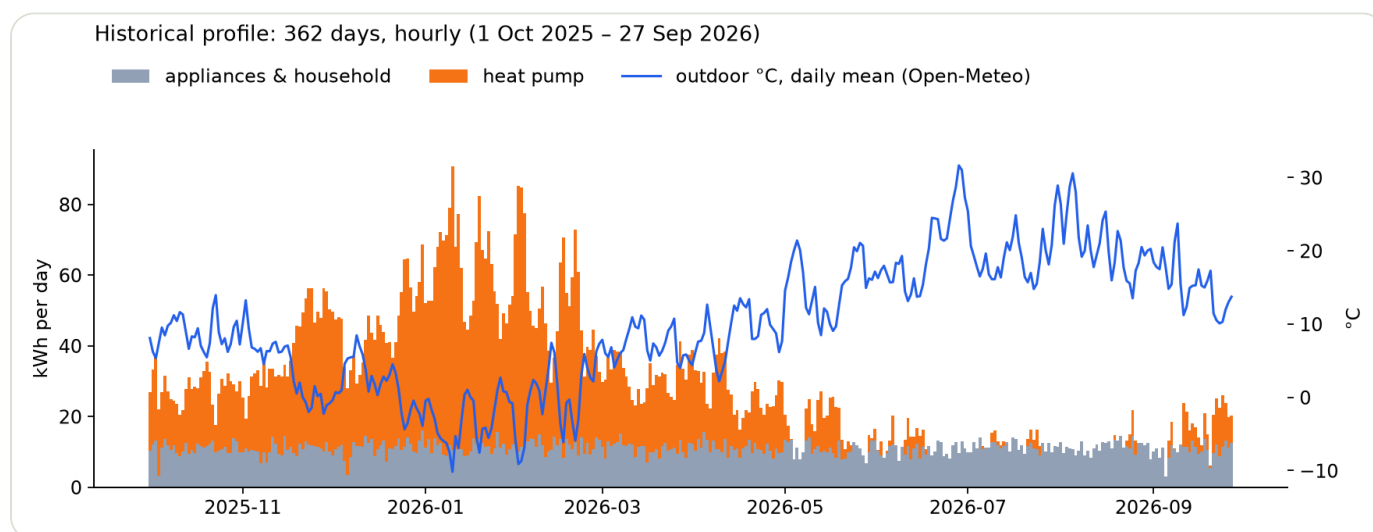


Figure 7. Daily consumption over 362 days: 10,663 kWh in total, of which the heat pump uses 6,661 kWh (62 %). Days range from 3 kWh (family away in summer) to 91 kWh (coldest January days).

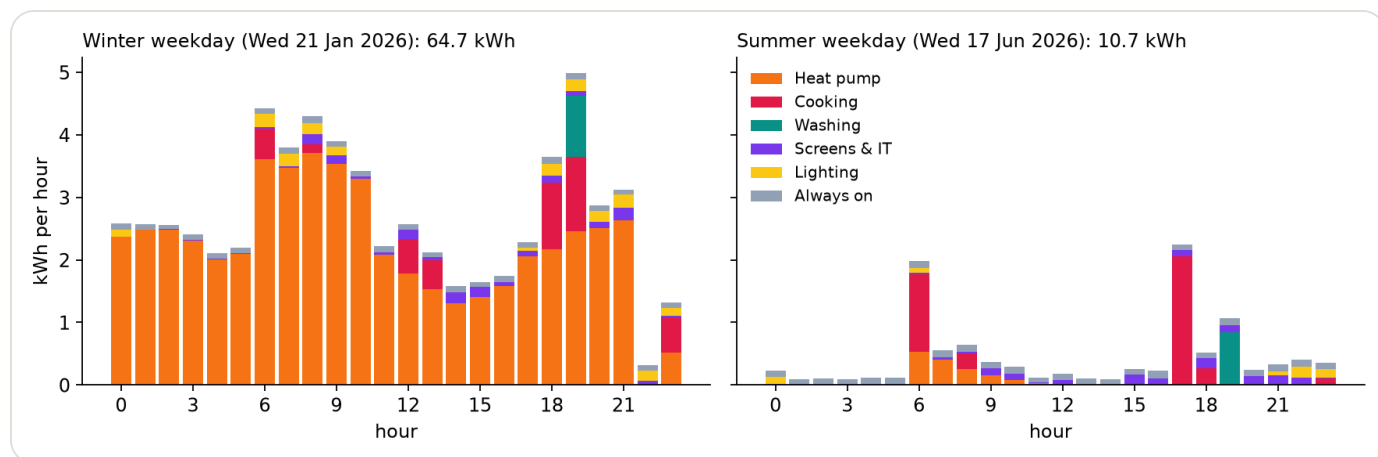


Figure 8. The same family on a winter and a summer weekday: heating dominates in January, while in June the morning and dinner peaks come from cooking, and a washing cycle runs after dinner.

Brief requirement for the profile	How it is met
Reflect the residents' daily routines	Hour-by-hour calendar following the brief's timetable (5:30–8:00, 8:00–14:00, 14:00–17:30, 17:30–21:30, night)
Include appliance and system operation	All 16 appliances and the heat pump, within the brief's power ranges
Account for outdoor temperature	Physical heat-pump model on real hourly weather
Realistic differences between days	Shifts, home-office days, meal times $\pm$ 45 min, cycles, outings
Periods when different members are at home	Presence per person and hour, visible in "Who's home"
Behavioural variability	Unplanned layer, simultaneous appliances creating short peaks

Eight models, one honest test

Inputs are only what is known before the hour: time of day, weekday, holidays and school days; the outdoor temperature, its 24-hour mean, heating degrees and solar radiation (from the forecast); and the **planned** calendar (who is expected home, awake or asleep, wake-ups, arrivals, Marek's shift, Ania's home office). Unplanned events are not inputs.

Test: a rolling-origin backtest on 8 weeks (December, January, February, April and the 4 most recent weeks). For each week every model is trained only on data before it and forecasts 7 days ahead with realistic weather-forecast errors added ($\approx 0.6\text{ }^{\circ}\text{C}$ on day 1 growing to $\approx 2\text{ }^{\circ}\text{C}$ on day 7).

Model	Hourly error	24 h total	3 days	7 days
Ensemble (XGBoost + gradient boosting) · chosen	0.31 kWh	11.1 %	3.5 %	3.7 %
XGBoost, direct	0.32 kWh	11.1 %	5.5 %	6.3 %
Gradient boosting (scikit-learn)	0.32 kWh	12.6 %	4.0 %	3.9 %
XGBoost, recursive (with lags)	0.33 kWh	12.0 %	5.2 %	5.0 %
GRU neural network	0.32 kWh	13.0 %	6.1 %	2.7 %
LSTM neural network	0.33 kWh	15.4 %	6.7 %	5.7 %
Ridge regression	0.46 kWh	21.3 %	11.2 %	8.0 %
Seasonal baseline (same hour last week)	0.68 kWh	42.4 %	37.4 %	35.5 %

The ensemble wins on the combined ranking (hourly error plus daily error, after calibration). With one year of hourly data and strong known inputs, tree models learn the patterns with less data, train in seconds and give steadier results than the recurrent networks, which usually need many households or several years.

Planning on the safe side

For a household, under-predicting is worse than over-predicting: a plan built on too little energy fails. The forecast is therefore multiplied by **1.056**, chosen so that at most 1 day in 5 is higher

than forecast (21 % in the backtest), and each hour and period gets a **likely range** (10–90 % of past errors).

Accuracy of the calibrated forecast	A single hour	24 h total	3 days	7 days
Average error	±28 %	±12 %	±7 %	±9 %

Single hours are uncertain by nature: a meal 30 minutes later moves energy to the next hour. Over a day these shifts cancel out, which is why the plan and the costs are based on totals and ranges. Most of the 3- and 7-day error is the deliberate +7 % safety margin.

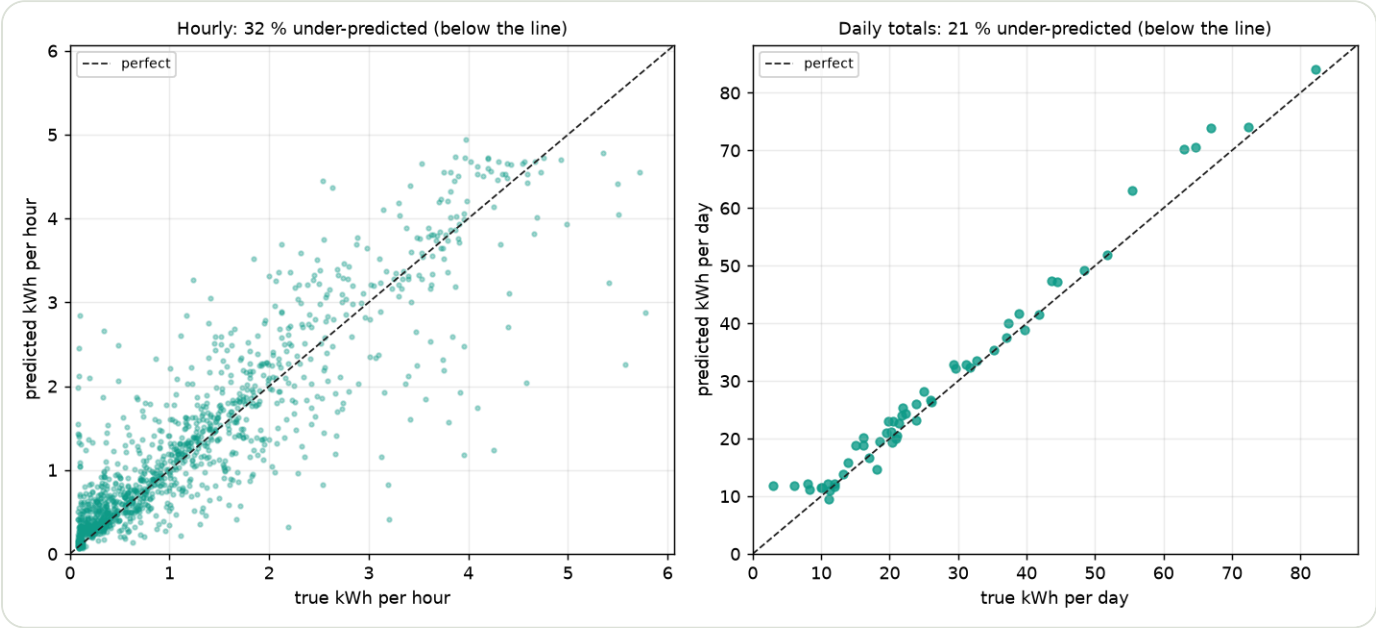


Figure 9. Backtest, true against predicted. Daily totals (right) sit close to the diagonal and mostly above it: the forecast errs on the safe side. All 8 test weeks were over-predicted in total (+4 % to +25 %), none under-predicted.

Short peaks and explanations

A second model (quantile XGBoost, 90 % level) forecasts the highest 1-minute power of each hour; on the last 4 weeks 88 % of hours stayed below it. The explanation text ("Why this forecast") is generated from the forecast itself: the total and temperature, the reason for each peak hour (grouped when they share a cause) and the heating share. Each day of the 7-day view also states why it is higher or lower than the average (shift, home office, weather).

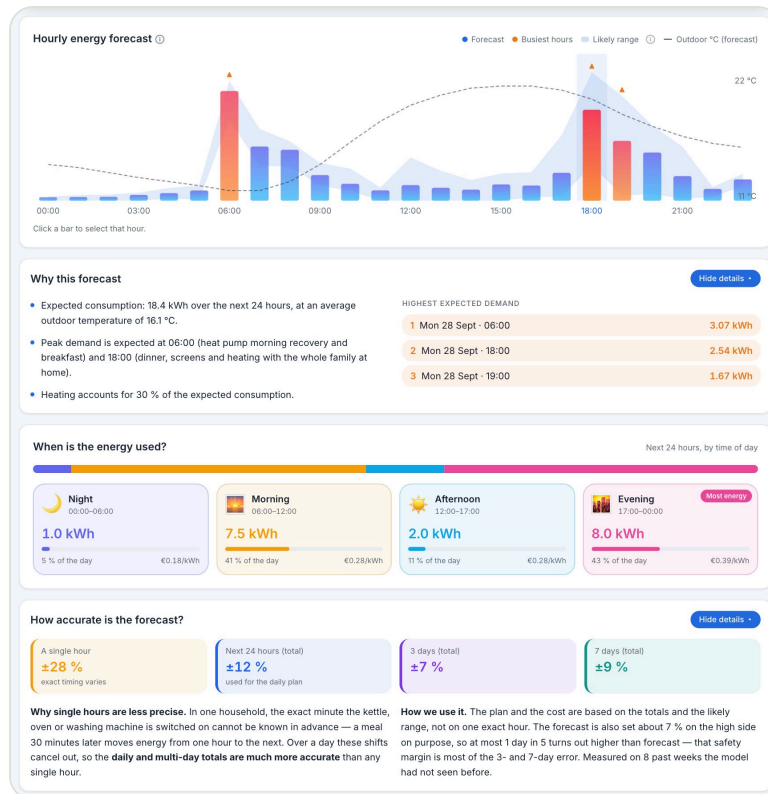


Figure 10. "Why this forecast" and "How accurate is the forecast?" explain the numbers in plain language.

Online and cumulative learning

When smart-meter readings arrive (`POST /api/readings`), every complete day is added to the history, the model is re-trained on all data and the calibration is refreshed (`update_model.py`). The correction reacts faster when reality is higher than forecast than when it is lower, so the safety margin never erodes. The backend also re-runs the pipeline by itself when the forecast is not for today.

7 • HOW IT WORKS: COST OPTIMISATION

The house as a heat battery

The optimiser (Google OR-Tools, a mixed-integer program) minimises the day's electricity cost under the HackoWatt tariff while keeping comfort:

- **Heat pump:** the indoor temperature must stay inside the chosen comfort band. The model knows the house's heat loss and thermal mass, so it can heat a little more at €0.18–0.28 and let the house coast through expensive hours.
- **Washing machine and dishwasher** may start anywhere inside their allowed windows (washing 04:00–17:00, dishwasher 21:00–07:00), where solar power or the night price make them cheapest.
- All other appliances stay where the family uses them: no lifestyle change.

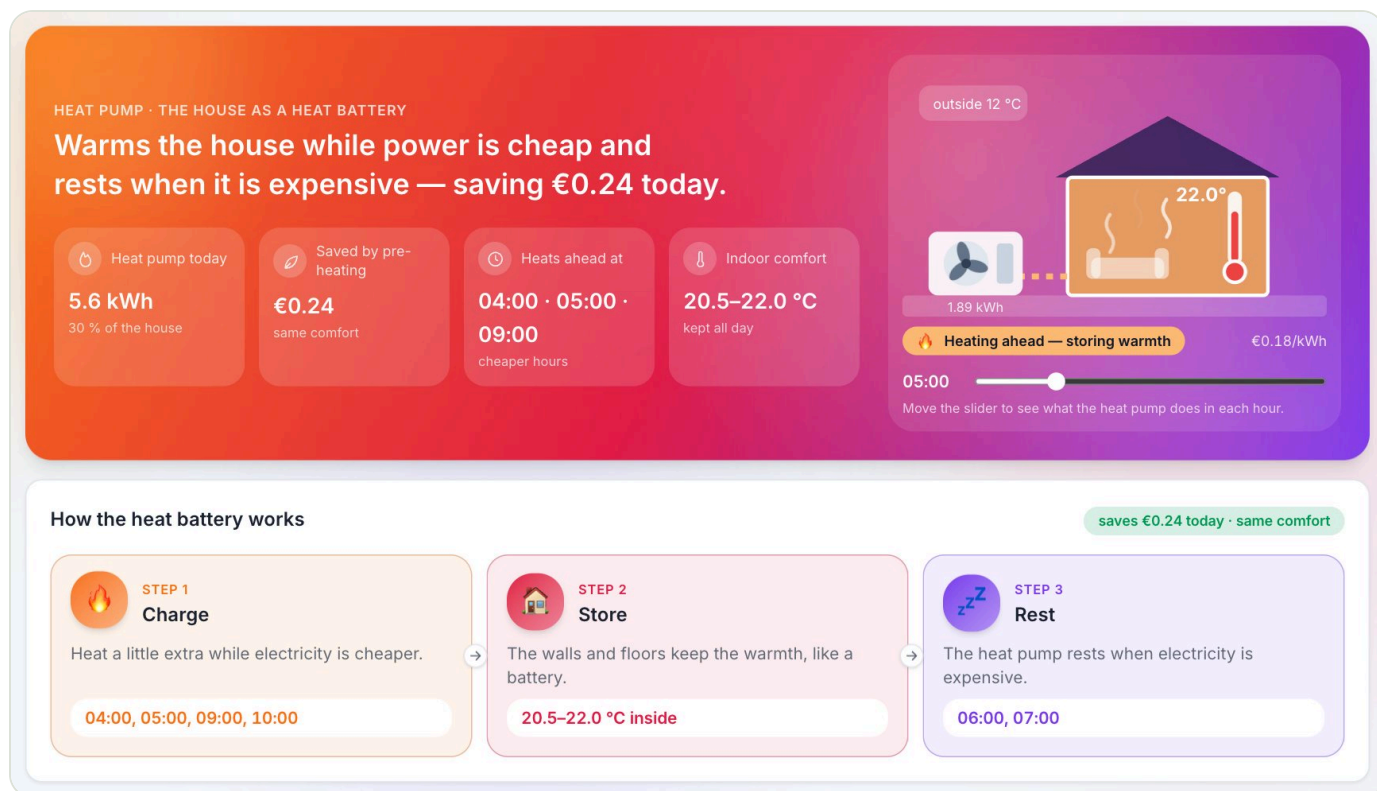


Figure 11. Heat pump page. At 05:00 the pump heats ahead at €0.18/kWh (indoor 22.0 °C); at 06:00–07:00 it rests on the stored warmth. The slider shows every hour.



Figure 12. Usual heating (grey) and optimised heating (orange) with the indoor temperature staying inside the comfort band.

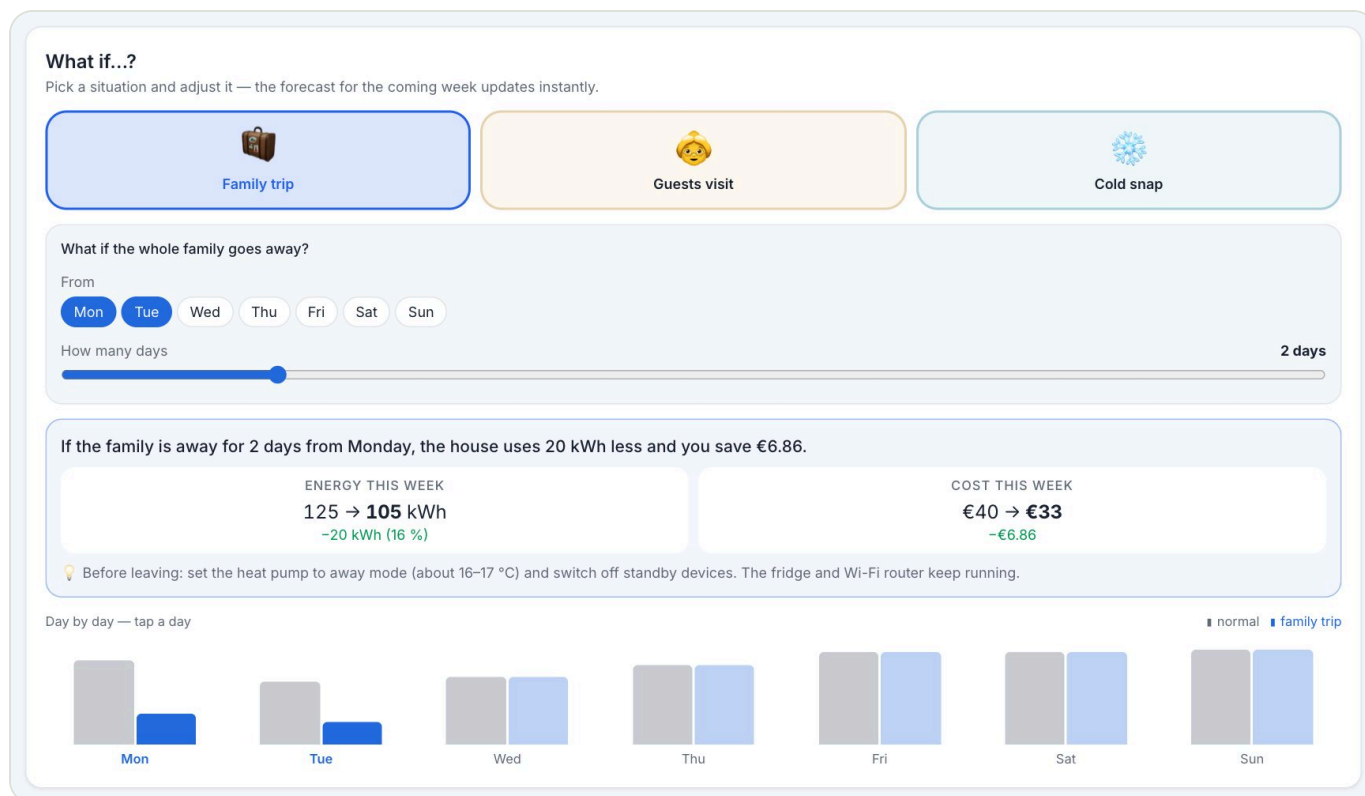


Figure 13. The what-if planner recalculates the week for a family trip, guests or a cold snap.

8 · HOW IT WORKS: RENEWABLE ENERGY SIMULATOR

Which roof pays off

Production comes from JRC PVGIS for Katowice (south-facing, 35° tilt, 14 % system loss, 2019–2023 hourly average: **1,072 kWh per kWp per year**). Every size is simulated hour by hour over a typical year against the household's demand, using the common assumptions: **€1,300 per kWp**, export **€0.08/kWh**, running cost **1 % of the investment per year**.

Net saving = lower bill + export income – running cost. Payback = investment ÷ net saving. **A** keeps current habits; **B** also moves the washing machine and dishwasher to the best hour of their window.

Size	Investment	Production	Demand covered	Less from grid	Net saving	Payback A	Payback B
3 kWp	€3,900	3,224 kWh	17 %	1,796 kWh	€595/yr	6.6 yrs	5.6 yrs
5 kWp	€6,500	5,374 kWh	21 %	2,296 kWh	€848/yr	7.7 yrs	6.8 yrs
8 kWp	€10,400	8,598 kWh	26 %	2,773 kWh	€1,171/yr	8.9 yrs	8.2 yrs
10 kWp	€13,000	10,748 kWh	28 %	2,996 kWh	€1,366/yr	9.5 yrs	8.9 yrs

Interpretation. Small systems pay back fastest because the house uses almost all their power at €0.18–0.40. Larger systems save more per year but send a growing share to the grid at only €0.08, so payback lengthens. Most of the family's demand is winter heating, when PV produces least, which caps the share covered below 30 %. Shifting appliances to sunny hours shortens payback by about one year for every size.

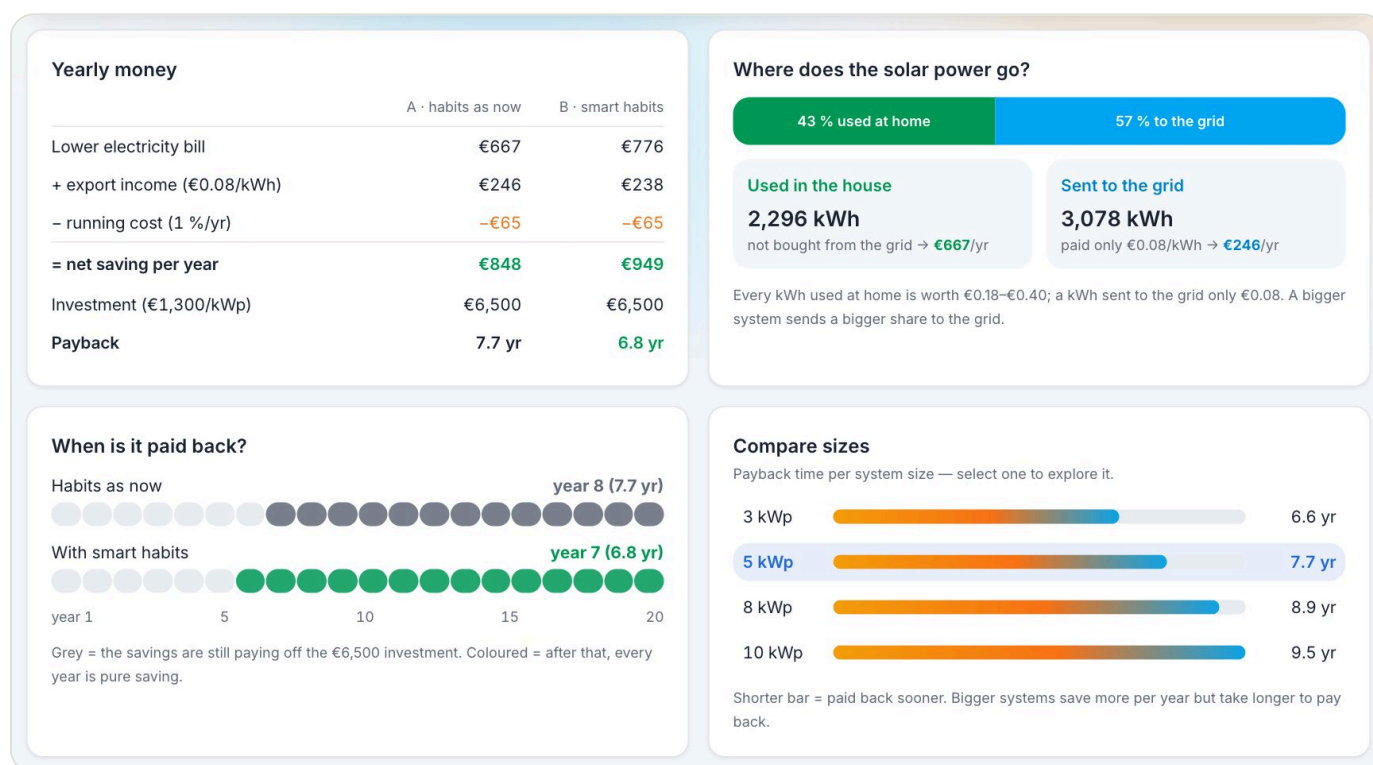


Figure 14. Yearly money for A and B, where the solar power goes (home or grid), when it is paid back, and a comparison of sizes.

One dashboard for the whole family

Six pages, each opening with its key numbers and a one-line explanation. Every figure has a reason next to it, and every chart is interactive (hover for values, click an hour to explain it).

Page	What the family sees
Overview	3D digital twin with clickable rooms, today at a glance, the four steps, "Right now" traffic light with a spoken summary, forecast chart, suggested habit changes, solar card, who's home, best times this week
Forecast	24 h / 3 days / 7 days, likely range, temperature, peak hours, why this forecast, accuracy, energy by time of day, day by day
Heat pump	Interactive heat-battery scene, charge → store → rest, comfort level slider, heating plan chart
Cost & plan	Usual vs plan, hourly cost chart over the tariff zones, what changes today, today's actions, what-if planner, everyday equivalents
Solar	Size slider 0–15 kWp, production, coverage, savings, payback A and B, size comparison
Appliances	Biggest users as bubbles, appliance share (hour, 24 h, 3 and 7 days), table per appliance and room

Also: **Ask EnergyPilot** (questions answered from the family's own forecast and plan, with Claude on the backend and an offline fallback), **personalised insights**, a **PDF and CSV report**, light and dark mode, and a **single HTML file** that runs offline with the data snapshot.

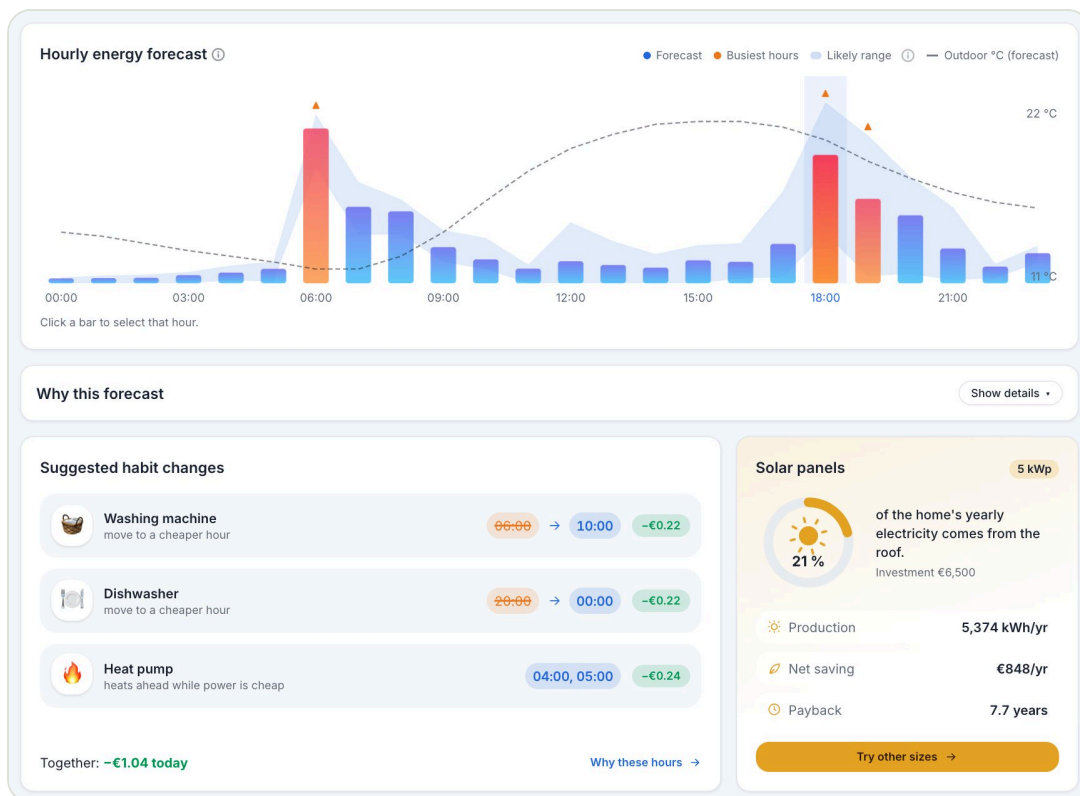


Figure 15. Overview: "Right now", the forecast and suggested habit changes as pictures (usual time → new time, saving).

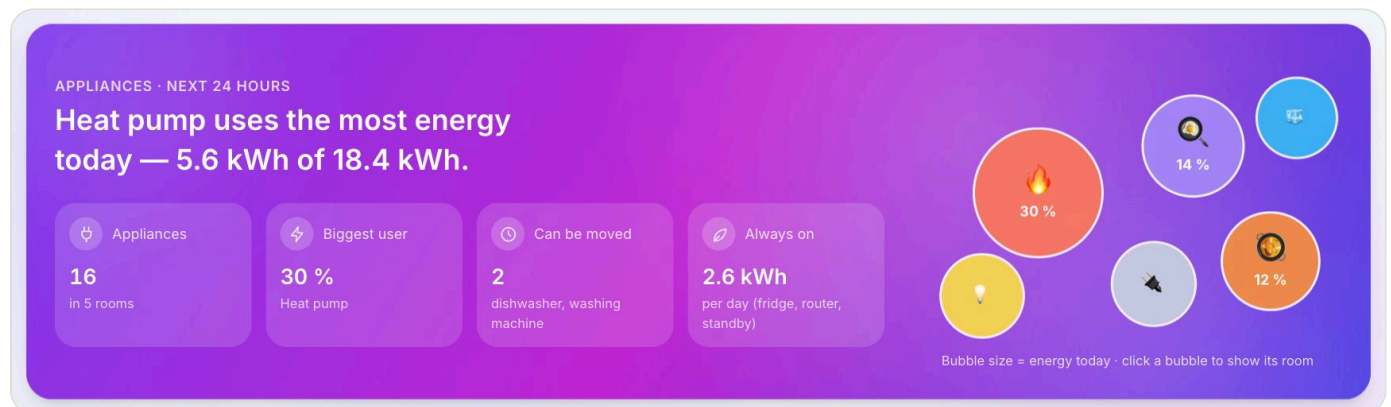
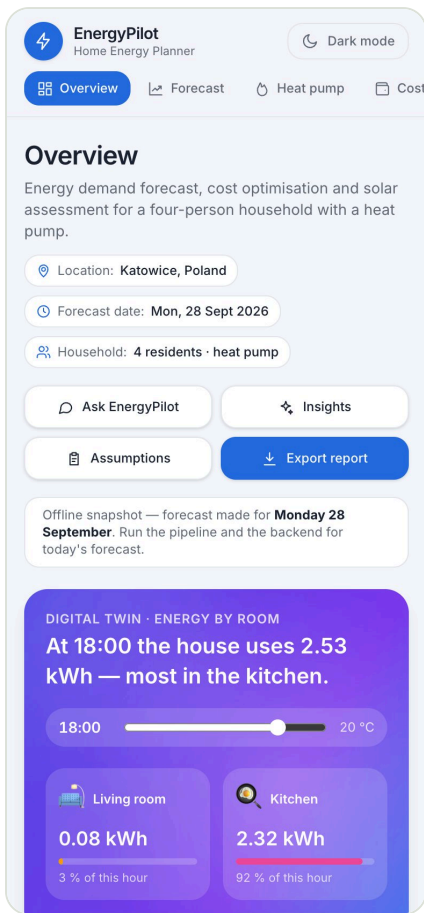
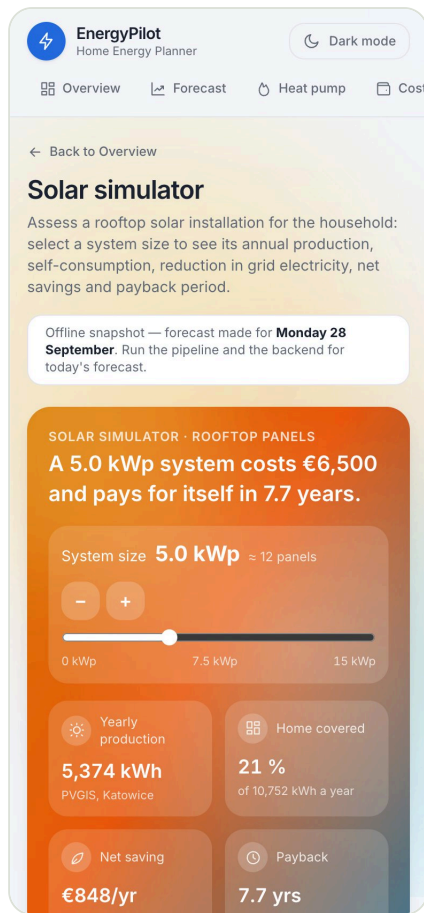


Figure 16. Appliances: the heat pump uses 30 % of today's energy; clicking a bubble filters the table to its room.

Designed for every screen



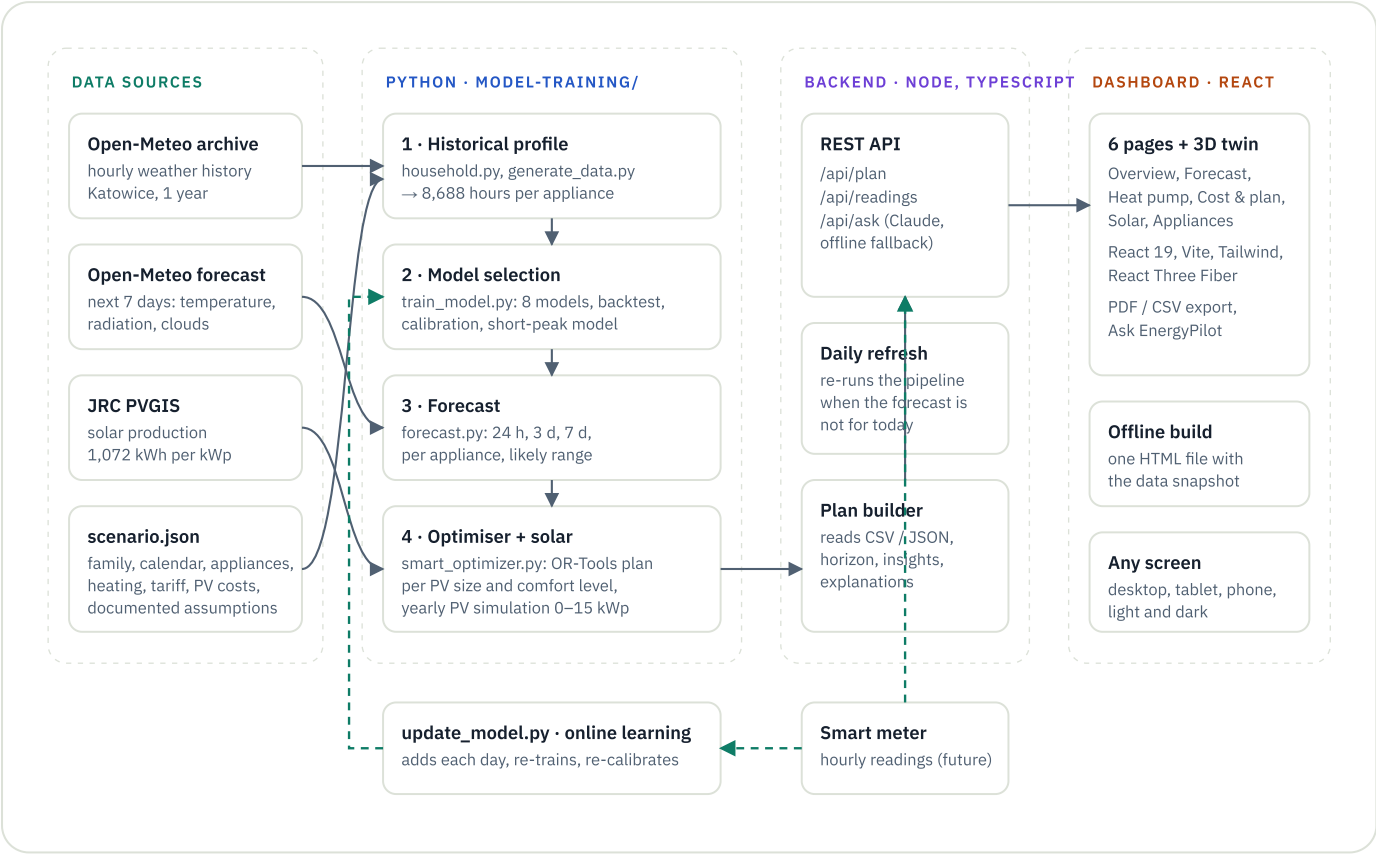
Overview on a phone



Solar on a phone

The layout adapts from desktop to tablet and phone with no sideways scrolling; the sidebar becomes swipeable tabs, and touch targets grow on touch screens. Animations respect the "reduce motion" setting.

From real weather to the family's screen



Each step writes plain files (`data/*.csv` , `model-training/output/*` , `model-training/models/*.pkl`), so every stage can be inspected and re-run on its own. The shared TypeScript contract (`shared/types.ts`) keeps the backend and dashboard in step, and the whole scenario (family, appliances, tariff) lives in one configuration file, so another household can be modelled without code changes.

Layer	Technology
Data and models	Python 3.12, pandas, NumPy, scikit-learn, XGBoost, PyTorch (LSTM, GRU), Google OR-Tools, matplotlib
Backend	Node.js, TypeScript (tsx), REST API, Anthropic Claude for "Ask EnergyPilot" (optional, with offline answers)
Frontend	React 19, Vite 7, Tailwind CSS 4, React Three Fiber (3D), jsPDF
External data	Open-Meteo (archive and forecast), JRC PVGIS

Innovation, usability, technology, reasoning

Innovation & creativity

30 pts

- **The house as a heat battery.** The optimiser uses the building's thermal mass as storage: it pre-heats in cheap hours and rests in expensive ones, moving heating energy out of the peak without buying a battery.
- **The family calendar as a sensor.** Marek's shift rota, Ania's home-office days and school drive the forecast; most demand forecasts use only time and weather.
- **Safe-side forecasting.** Calibrated so the family is rarely caught short, with a likely range, and an online correction that reacts faster upwards than downwards.
- **Solar judged together with habits.** Payback A and B side by side links behaviour change and investment in one answer.
- **Development potential:** the scenario is configuration, the readings API and online learning are ready for a real smart meter, and dynamic tariffs, a home battery or an EV fit into the same optimiser.

Functionality & usability

30 pts

- **User value:** concrete actions with their saving (washing machine at 10:00, –€0.22), a 23 % lower bill at the same comfort, and a clear solar decision.
- **Intuitive:** four steps on the first page, one headline per page, plain-language reasons next to every number, a traffic light for "Is now a good time?".
- **Interactive:** hour sliders, a clickable 3D house, comfort and PV-size sliders, what-if scenarios, clickable charts.
- **Effective everywhere:** desktop, tablet and phone, light and dark, offline single file, PDF and CSV export, and a question assistant.

Technical implementation

20 pts

- **Model performance:** 8 candidates compared in a rolling backtest with weather-forecast noise; $\pm 12\%$ on the 24-hour total, $\pm 7\text{--}9\%$ over 3–7 days; no test week under-predicted.
- **Optimisation:** a mixed-integer program with a thermal model of the house, solved for every PV size and comfort level.
- **Architecture:** a reproducible pipeline (one command), file-based stages, a typed contract between backend and frontend, a daily automatic refresh and online learning.
- **Code quality:** small single-purpose modules, one configuration file, documented assumptions and a complete README; the frontend type-checks cleanly.
- **UI:** a consistent design system (tokens for colours and type), accessible contrast, reduced-motion support and responsive layouts.

Presentation & justification

20 pts

- Every design choice is justified in this document: why a simulated profile (required by the brief), why this model (backtest), why a safe-side forecast (the cost of under-predicting), why the heat battery (the heat pump is 62 % of demand).
- All four questions are answered with numbers from the running system (section 3), and each required outcome is mapped to where it lives (section 4).
- Limitations are stated openly (section 13), with concrete next steps.

Everything we assumed, in one place

All assumptions are stored in `model-training/scenario.json` and shown in the dashboard under **Assumptions**.

Household and schedules

- Marek works a 3-shift rota that rotates every week: morning (06–14, away 06–15), afternoon (14–22, away 14–23), night (22–06, away 22–07, sleeps 07–14), Monday to Friday, plus a Saturday shift on 1 in 4 weekends.
- Ania works from home every Tuesday and Thursday and on 25 % of the other weekdays; on office days she is away 08–17.
- Kuba (14) is at school 08–15 and Zosia (10) 08–14 on school days. Kuba has football Monday and Wednesday 16–18, Zosia dance Tuesday and Friday 16–18.
- School breaks: 22–31 Dec, 2–15 Feb (Silesian winter break), 2–7 Apr (Easter), 27 Jun–31 Aug. Polish public holidays including 24 Dec.
- Unplanned events the forecast cannot know: weekend outings (35 % of weekend days, 3–6 h), evenings out, sick days, occasional weekend trips.

Appliances

- Power and energy per use are drawn uniformly from the ranges in the brief (for example kettle $2\text{ kW} \times 3\text{--}5\text{ min}$, hob $1.5\text{--}3.5\text{ kW}$).
- Cooking follows meals: breakfast at each person's wake-up time, lunch when an adult is home, the children's snack after school, dinner 17:30–19:30 ($\pm 45\text{ min}$), and a late meal after Marek's shift.
- Screens run only when their users are at home and awake; Ania's laptop runs through her home-office hours.
- Washing machine ≈ 4.5 cycles a week, dishwasher ≈ 6 cycles a week, both started only when an adult is home.
- Lighting is on when people are awake at home and the real solar radiation is below 20 W/m^2 .

Heating

- Air-to-water heat pump; older house with a heat loss of 0.25 kW per $^{\circ}\text{C}$. COP falls from 3.6 at 7°C to 2.4 at -7°C .
- Setpoints 21°C by day, 19.5°C at night (22–06), 20°C when nobody is home. Heating fades out as the 24-hour mean outdoor temperature rises from 13°C (full) to 17°C (off).

- Electric power 1.5–3.5 kW while running (brief); a smaller need means the pump cycles within the hour. Internal gains 80 W per person; solar gains from real radiation. Domestic hot water is not included (not in the brief).

Weather, tariff and solar

- Real hourly weather for Katowice (50.26 °N, 19.02 °E) from Open-Meteo: archive for the history, forecast API for the next 7 days. Local time (Europe/Warsaw); daylight-saving hours are interpolated so every day has 24 hours.
- HackoWatt common tariff: €0.18 (00–06), €0.28 (06–17), €0.40 (17–22), €0.28 (22–24) per kWh; export €0.08/kWh; PV €1,300 per kWp; running cost 1 % of the investment per year.
- PV from JRC PVGIS (south, 35° tilt, 14 % loss, 2019–2023): 1,072 kWh per kWp per year. Next-days PV from the Open-Meteo radiation forecast on the same tilted plane.

Forecast and optimisation

- The forecast knows the planned calendar and the weather forecast, not the unplanned events.
- The optimiser moves only the washing machine, the dishwasher and heat-pump timing; comfort stays inside the chosen band.
- No real smart meter is connected in this demo; the readings API and online learning are tested with a meter simulator.

13 · LIMITATIONS AND NEXT STEPS

What we would do next

LIMITATION

Simulated profile

As the brief requires, the history is simulated. We reduce the risk of the model learning our own rules: unplanned events are hidden from it and the test adds weather-forecast errors.

NEXT STEP

Connect a real meter

The readings API and online learning already exist; with real readings the simulated history is replaced day by day.

LIMITATION

Single hours are uncertain

$\pm 28\%$ per hour against $\pm 12\%$ per day. The plan uses totals and ranges; the short-peak model covers the minutes.

NEXT STEP

Dynamic prices and CO₂

Use hourly market prices and grid CO₂ intensity instead of the fixed bands, so the plan can also choose the cleanest hours.

LIMITATION

Fixed tariff and no hot water

We follow the common tariff and the brief's appliance list; domestic hot water would add another flexible load.

NEXT STEP

Battery and electric car

Both fit into the same optimiser as extra storage and flexible loads, and into the solar simulator's payback.

14 · HOW TO RUN IT

One command for the data, two for the app

1. Pipeline (weather, history, models, forecast, plan): `cd model-training && .venv/bin/python run_pipeline.py` (about 3 minutes; `--quick` skips the model comparison).
2. Backend: `npm run dev:backend` from the project folder (port 8787).
3. Dashboard: `npm run dev:frontend`, then open `http://localhost:5173`.
4. Offline: open `html/energypilot_family_dashboard.html` directly in any modern browser.

Folder	Contents
<code>docs/</code>	This document and the challenge brief
<code>html/</code>	The whole dashboard in one file
<code>data/</code>	Historical profile, weather, PVGIS data, offline snapshot
<code>model-training/</code>	Python: profile, model comparison and training, forecast, optimiser, trained model and results
<code>backend/</code>	Node + TypeScript API
<code>frontend/</code>	React dashboard
<code>shared/</code>	Data contract used by backend and frontend

The README in the project folder documents every file, every model and every setting in detail.

Data sources: Open-Meteo Historical Weather API and Forecast API (open-meteo.com), JRC PVGIS (European Commission), HackoWatt 2026 Scenario 2 brief and Common Challenge Assumptions.

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